

South-facing courtyard, one connected interior

Working location: Northumberland | Plan 1:100 at A3, printed at 100% | South is at the top; north at the bottom



S1 / SHADE THE ROOF FIRST

Retractable external roof shade below the high vent strip.
Retain insulating solar-control glass throughout.
The unshaded upper strip needs explicit solar testing.

S2 + S3 / SHADE THE VERTICAL GLASS

Reserve external screens at the south dining / garden sliders (S2) and east-facing garden-room side window (S3).
Retain the current opening widths and bronze frames.

V1 - V3 / A ROUTE FOR EXCESS HEAT

V1: attended south door opening; secure intake to design.
V2: two high roof vent reservations, subject to pitch / system.
V3: opening portions in the north dining window.
Keep widths fixed; calculate effective opening areas.

H1 / HEAT THE CONNECTED SPACE

Develop wet underfloor heating with a heat-pump design.
Coordinate garden / shared-room loops and controls.
Exclude fixed joinery and hearth; size from heat loss.
Check comfort by glass before deciding on extra emitters.

FRESH AIR AND SERVICES

Study whole-house MVHR separately from summer purge.
Reserve accessible duct routes in service / hall ceilings.
Plant, terminals and manifolds have no selected locations.
Keep the vault, lighting ledges and roof head coordinated.

ORIENTATION CHANGES THE CHECKS

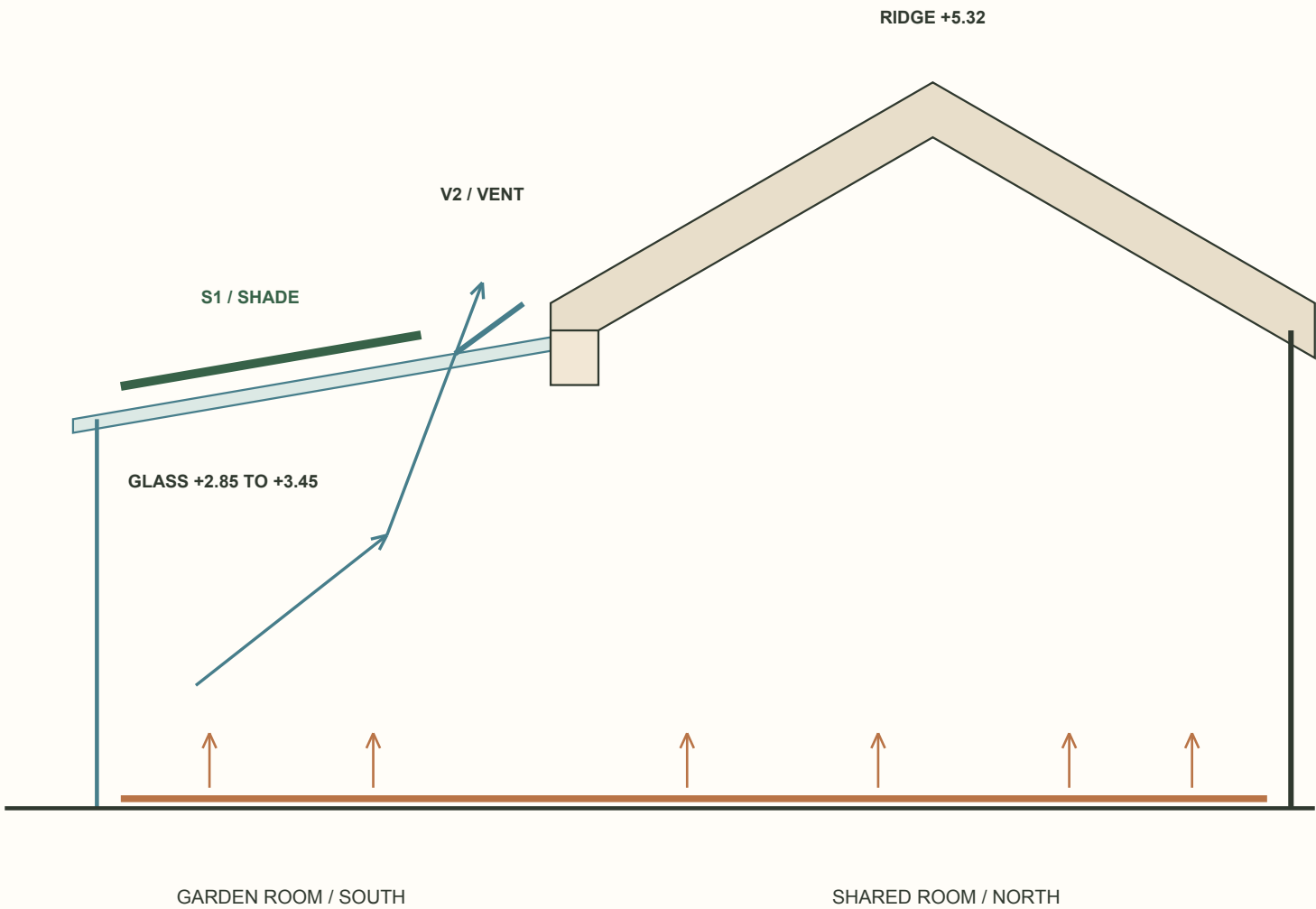
South-facing roof and front glass: priority solar assessment.
West-facing family windows: check afternoon bedroom gains.
Courtyard shelter, trees and neighbouring shadows unknown.

SITE ASSUMPTION

Northumberland; due-south courtyard axis for this study.
Exact plot, exposure, altitude and weather file remain open.

Give the shade and vents their own space

Section A-A at plan x = 13.20 m | 1:50 at A3 | South / courtyard left; north / rear right | Heights retained from concept 07B



THE JUNCTION TO RESOLVE

Stop the shade at y = 11.85 m in this reservation.
Keep roof vents in the upper strip at y = 12.10-12.60 m.
This avoids overlap in plan, not a proven assembly fit.
The remaining upper glass strip is approximately 0.95 m.

WHAT THE SUPPLIERS MUST COORDINATE

Actual roof pitch, vent upstands and weathering.
Shade cassette, guide rails, fixings and maintenance access.
Opening vent sweep and wind / rain controls.
Drainage below the main eave and around the vents.
A compatible system may require a revised junction.

KEEP THE INTERIOR INTENT

Retain the 3.10 m opening head and continuous floor.
Run services within planned hall / service ceiling zones.
Keep ambient lighting ledges and drivers accessible.
Do not fill the vault with ad hoc ducts or blind boxes.

VENTILATION IS A DESIGN RESERVATION

No opening free area or exhaust capacity is established.
Select a low-pitch-compatible system and test it together
with the external shade and secure air inlet.

GLASS + FRAME

Specify whole-unit insulation and solar performance.
Check visible light, neutral appearance and glare.
Review edge temperatures, airtightness and
thermal bridges at the roof / frame junction.

FLOOR + HEATING

Keep insulation continuous at the glazed threshold.
Check usable heated floor area and floor finish.
Use coordinated controls for the open interior.
The stove is an optional local heat source;
its output must suit the calculated room demand.

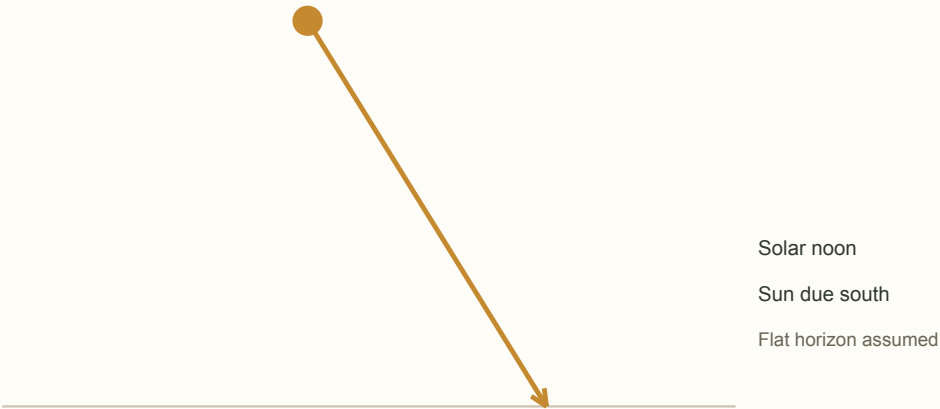
IF PASSIVE MEASURES FALL SHORT

Test shading and usable ventilation first.
If modelling still shows overheating, investigate
targeted active cooling with a services designer.
No cooling capacity or location is selected.

Let the house behave differently by season

Operating intent and a simple solar-noon comparison | Illustrations are not a weather simulation or a prediction of indoor temperature

SUMMER SOLSTICE / APPROX. 58.2 DEGREES



WINTER SOLSTICE / APPROX. 11.4 DEGREES



SUNNY SUMMER DAY

Shade deployed

↓

Fresh air maintained

↓

Purge if outside is cooler

COOLER SUMMER NIGHT

Secure intake + high vent

↓

Purge only while cooler

↓

Rain / wind / security control

WINTER DAY

Shade retracted as useful

↓

Heat recovery + steady heating

↓

Purge vents normally closed

Angles use a representative latitude of 55.2 N: altitude = 90 - latitude + solar declination. Equinox: 34.8 degrees.

No plot coordinates, horizon, shade duration or wind direction are inferred. Closed-door night use needs a designed secure inlet.

NEXT TECHNICAL CHECK

- Model the connected interior and the bedrooms using site-appropriate current / future weather.
- Test glazing, shade, usable openings, occupancy and heat gains; calculate room-by-room winter heat loss.
- Select opening areas, heating output and control settings from those results. Confirm the applicable regulations.

References supporting the design brief; no compliance assessment has been carried out:

CIBSE: year-round comfort and overheating

GOV.UK: Overheating (Document O)

Energy Saving Trust: wet underfloor heating

Ventilation guidance (Document F)